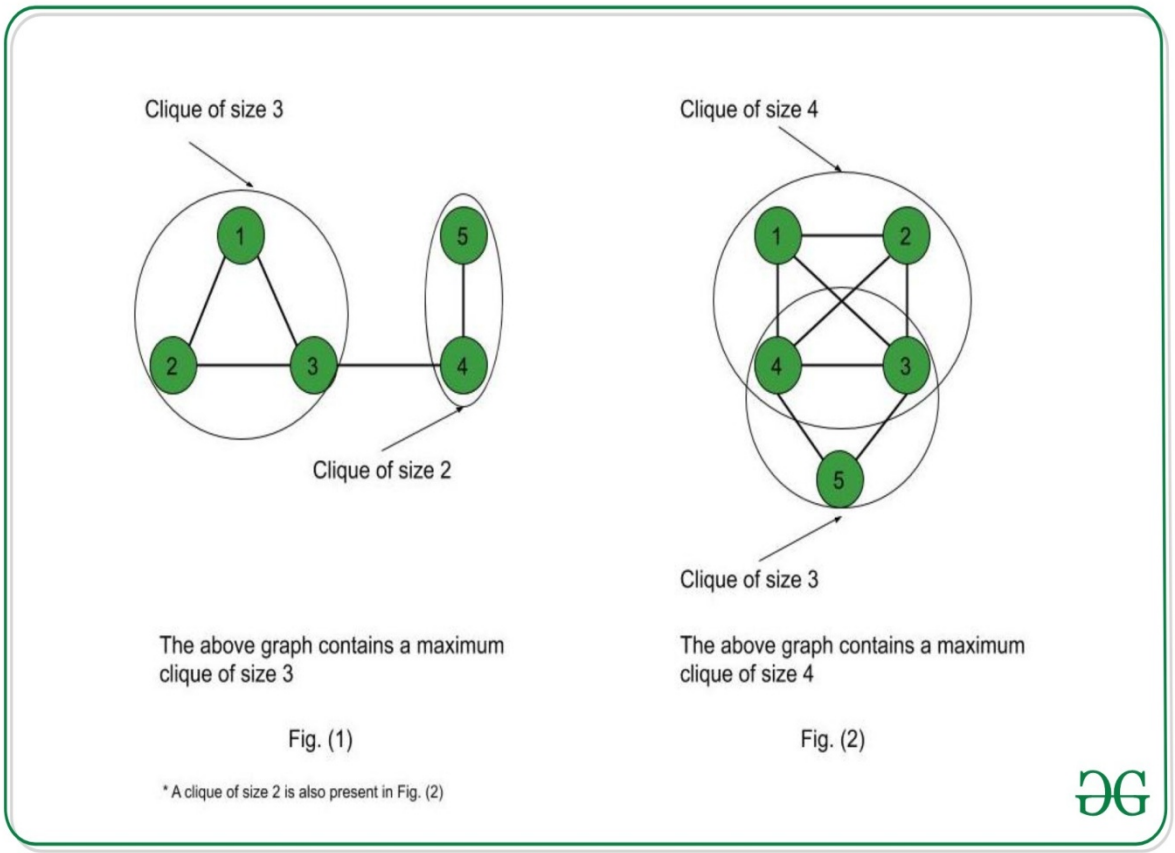
CLIQUE PROBLEM

- A clique is a sub graph of a graph such that all the vertices in this sub graph are connected with each other that is the sub graph is a complete graph.
- The Maximal Clique Problem is to find the maximum sized clique of a given graph G that is a complete graph which is a sub graph of G and contains the maximum number of vertices.
- This is an optimization problem. (Finding maximum clique of the graph)
- Correspondingly, the Clique Decision Problem is to find if a clique of size k exists in the given graph or not.



- To prove that a problem is NP-Complete, we have to show that it belongs to both NP and NP-Hard Classes.

The Clique Decision Problem belongs to NP

- If a problem belongs to the NP class, then it should have polynomial-time verifiability that is given a certificate, we should be able to verify in polynomial time if it is a solution to the problem.

Proof:

1. **Certificate** – Let the certificate be a set S consisting of nodes in the clique and S is a sub graph of G.
2. **Verification** – We have to check if there exists a clique of size k in the graph. Hence, verifying if number of nodes in S equals k.

- To check if the graph formed by the k nodes in S is complete or not, it takes $O(k^2) = O(n^2)$ time
- Therefore, the Clique Decision Problem has polynomial time verifiability and hence belongs to the NP Class.

The Clique Decision Problem belongs to NP-Hard

- A problem L belongs to NP-Hard if every NP problem is reducible to L in polynomial time.
- Now, let the Clique Decision Problem by C.
- To prove that C is NP-Hard, we take an already known NP-Hard problem, say S, and reduce it to C for a particular instance.

- If this reduction can be done in polynomial time, then C is also an NP-Hard problem.

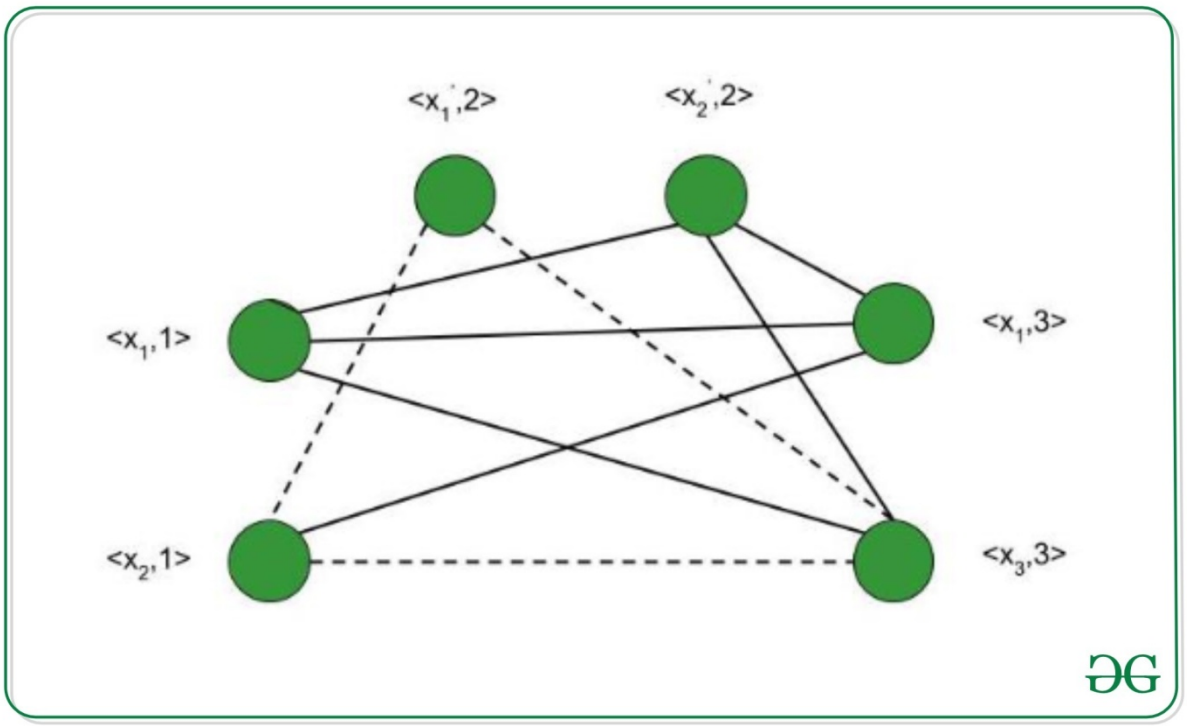
The Boolean Satisfiability Problem (S) is an NP-Complete problem

Proof that the Boolean Satisfiability problem reduces to the Clique Decision Problem

- ✓ Let the Boolean expression be – F = (x1 v x2) ^ (x1' v x2') ^ (x1 v x3) where x1, x2, x3 are the variables, '^' denotes logical 'and', 'v' denotes logical 'or' and x' denotes the complement of x.
- ✓ Let the expression within each parentheses be a clause.
- ✓ Hence we have three clauses – C1, C2 and C3.
- ✓ Consider the vertices as – <x1, 1>; <x2, 1>; <x1', 2>; <x2', 2>; <x1, 3>; <x3, 3> where the second term in each vertex denotes the clause number they belong to.

We connect these vertices such that

1. No two vertices belonging to the same clause are connected.
2. No variable is connected to its complement.



- Thus, the graph G (V, E) is constructed such that
- V = { <a, i> | a belongs to Ci } and E = { (<a, i>, <b, j>) | i is not equal to j ; b is not equal to a' }
- Consider the subgraph of G with the vertices <x2, 1>; <x1', 2>; <x3, 3>.
- It forms a clique of size 3 (Depicted by dotted line in above figure) . Corresponding to this, for the assignment – <x1, x2, x3> = <0, 1, 1> F evaluates to true.
- Therefore, if we have k clauses in our satisfiability expression, we get a max clique of size k and for the corresponding assignment of values, the satisfiability expression evaluates to true.
- Hence, for a particular instance, the satisfiability problem is reduced to the clique decision problem.
- Therefore, the Clique Decision Problem is NP-Hard.

Conclusion

- ✓ The Clique Decision Problem is NP and NP-Hard.
- ✓ Therefore, the Clique decision problem is NP-Complete.